

Himanshu Hemant Paithane

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PROFESSIONAL SUMMARY

Software and data engineer with experience building automation tooling, scalable data pipelines, ML workflows, and data-intensive applications across startups, AI SaaS, and Fortune 500 environments. Experienced in Python, TypeScript/JavaScript, SQL, PyTorch, computer vision, distributed data processing, CI/CD, and cloud systems, with hands-on experience developing real-time ML applications and reliable data systems supporting production analytics and AI use cases.

SKILLS

- **Programming:** Python, SQL, TypeScript, JavaScript, React, C++, Bash, PySpark
- **Machine Learning:** PyTorch, YOLO, OpenCV, Computer Vision, TensorFlow, scikit-learn, XGBoost, LightGBM, Model Evaluation, A/B Testing
- **Data Engineering:** Apache Spark, ETL/ELT, Data Pipelines, Data Modeling, Data Quality, Workflow Orchestration, Airflow, Kafka, Databricks, BigQuery, MongoDB
- **AI Engineering:** LLMs, RAG, Embeddings, Vector Search, ChromaDB, LangChain, LangGraph, MCP, Agentic Workflows
- **Cloud & DevOps:** GCP, AWS, Azure, Docker, Terraform, CI/CD, Databricks

WORK EXPERIENCE

Software Engineer, Data | FiPet (Early-Stage Startup)

Aug 2026 - Present

- Developing TypeScript-based web features and internal tools for a gamified financial literacy platform, iteratively shipping user-facing functionality and workflow improvements based on product and operational requirements.
- Developing efficient Python-based automation workflows and data pipelines to securely process and store sensitive user activity data for a gamified financial literacy platform, enabling user engagement analytics, automated reporting, as well as driving data-driven product and operational decisions.

Data Engineer, ML Analytics | Panasonic North America

May 2025 - May 2026

- Developed and deployed a **Databricks** Machine Learning analytics pipeline to transform and reconcile **300,000+** customer records using a **PyTorch**, **NLP** and fuzzy matching algorithm to improve customer segmentation, data quality, and the reliability of downstream analytics and ML use cases.
- Built a scalable **SQL** and **PySpark** workflow to automate data quality, observability, schema checks and **ETL** reconciliation, improving pipeline reliability and ensuring trusted datasets for **CI/CD** pipelines, analytics products, dashboards and machine learning models.
- Developed and maintained Bronze, Silver, and Gold data layers in Microsoft Fabric using Dataflows and PySpark notebooks, implementing ETL, data cleansing, and transformation workflows to deliver reliable analytics-ready datasets.

Data Scientist | Jar (Fintech Startup, ~\$500M Valuation)

Jan 2024 - Aug 2024

- Developed, validated and deployed an **ML**-based affluence scoring model using **LightGBM** and **XGBoost** to classify new users into spending-affinity tiers, improving customer targeting decisions while replacing a third-party solution that saved **~\$15k/month**.
- Built **Airflow**-orchestrated **ETL** pipelines to process high-volume customer transaction events generated by **~20k** daily active users streamed through **Kafka**, transforming raw data stored in **AWS S3** into curated **MongoDB** datasets that powered analytics, KPI reporting, product insights and machine learning models.
- Applied **NLP** techniques to transform **terabytes** of unstructured financial transaction SMS data into structured datasets for user segmentation, behavioral analysis, spending-trend identification, and predictive modeling, with insights delivered through interactive **Tableau** dashboards.

Data Science Intern | SymphonyAI (AI SaaS)

May 2023 - Aug 2023

- Built and optimized machine learning models for **time-series forecasting** and pattern recognition of motor health to identify exceptions and risk factors using **sensor-derived time-series data**, improving accuracy by **23%** through feature engineering, hyperparameter tuning, performance analysis and model evaluation.
- Improved accuracy of production ML models including **Regression** based models and **neural networks** through feature engineering, statistical analysis, and hyperparameter tuning, boosting **F1-score** to **91%** and enabling more reliable predictions at scale.
- Built end-to-end data preprocessing, feature engineering, model training, and validation workflows across diverse ML models and pipelines to deliver reliable, production-oriented predictions and performance analysis.

Data Science Intern | Sharp

Jun 2021 - Sep 2021

- Built production-ready ML and analytical workflows using Python, PyTorch, scikit-learn, and TensorFlow, conducted **A/B testing** to inform product decisions, and optimized Spark/Hadoop data pipelines to improve large-scale processing efficiency by 4x

PROJECTS (Check out more [here](#))

Wake Bot | Computer Vision, PyTorch, YOLO, OpenCV, Labelling | [GitHub Link](#)

- Built and annotated a computer vision training dataset using Labellmg, labeling facial features, eye states, and head postures to create YOLO-compatible training data for drowsiness detection.
- Fine-tuned YOLO models using PyTorch on labeled image data to detect visual indicators of driver drowsiness, combing eye state, facial features and head position for real time classification
- Developed an OpenCV-based inference pipeline for real-time video capture, preprocessing, and model execution, optimizing the end-to-end workflow for detection.

AgenticRAG | Python, OpenAI Function Calling, RAG, Vector Search, ChromaDB, Embeddings | [GitHub Link](#)

- Built an end-to-end agentic RAG application that uses OpenAI function calling and local vector retrieval to answer Python programming questions with grounded context from official documentation.
- Constructed a local documentation knowledge base using token-based chunking and embeddings to support semantic retrieval across tutorials, reference material, and concept explanations.
- Designed and orchestrated multi-step retrieval workflows combining semantic search, documentation lookup, and context selection to improve response relevance, reduce hallucinations and increase reliability of technical answers.

StructuraAI | Python, OpenAI API, PyMuPDF, JSON Schema, Document Intelligence Analytics | [GitHub Link](#)

- Built a document intelligence pipeline to extract structured metadata and schema definitions from financial document PDFs into machine-readable JSON outputs.
- Designed parsing logic to capture table layouts, field names, data formats, lengths, and metadata from semi-structured documents and transform them into structured records for downstream analytics workflows.
- Automated document-to-data conversion to support scalable reference-data extraction, improve consistency, and reduce manual processing effort.

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY

Master of Science, Computer Science (GPA: 3.9/4.0)

May 2026